NAME: _____ Geometry (FL B.E.S.T.)

3.2 Reflections

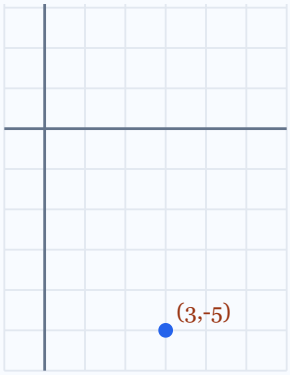
A reflection flips a figure over a mirror line so that every point and its image are the same distance from the line, on opposite sides — which is why reflections preserve distance but reverse orientation, unlike translations. Beyond the four memorized rules (x-axis, y-axis, $y = x$, $y = -x$), this section covers reflecting over any horizontal or vertical line using a distance formula, proving algebraically why distance survives while orientation flips, and working backward to find the mirror line itself from a point and its image.

Examples

Example 1: Reflect over the x-axis

Reflect $(3, 5)$ over the x-axis.

Solution



Flipping over the x-axis keeps x the same and negates y.

Reflecting over the x-axis keeps x the same and negates y — the point flips vertically, landing the same distance below the axis as it started above.

$(3, 5) \to (3, -5)$.

Answer: $(3, -5)$.

Example 2: Reflect over the y-axis

Reflect $(3, 5)$ over the y-axis.

Solution

Reflecting over the y-axis keeps y the same and negates x.

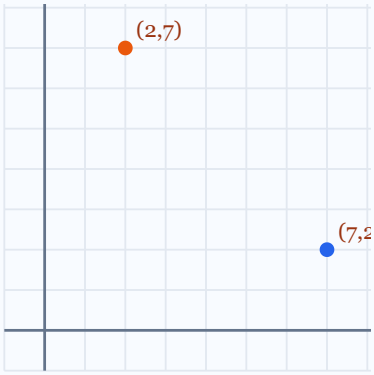
$(3, 5) \to (-3, 5)$.

Answer: $(-3, 5)$.

Example 3: Reflect over $y = x$

Reflect $(2, 7)$ over $y = x$.

Solution



Reflecting over y = x swaps the coordinates.

The mirror line $y = x$ treats horizontal and vertical distance symmetrically, so reflecting over it swaps the coordinates.

$(2, 7) \to (7, 2)$.

Answer: $(7, 2)$.

Example 4: Reflect over $y = -x$

Reflect $(4, 1)$ over $y = -x$.

Solution

Reflecting over $y = -x$ swaps the coordinates *and* negates both (the extra negation accounts for the line's negative slope).

$(4, 1) \to (-1, -4)$.

Answer: $(-1, -4)$.

Example 5: Reflect over a vertical line

Reflect $(1, 3)$ over the line $x = 2$.

Solution

For a vertical mirror at $x = k$, y stays the same, and the new x-coordinate is found with $x' = 2k - x$ (the image is as far past k as the original was before it).

$x' = 2(2) - 1 = 3$.

Answer: $(3, 3)$.

Example 6: Reflect over a horizontal line

Reflect $(5, 1)$ over the line $y = 3$.

Solution

For a horizontal mirror at $y = k$, x stays the same, and $y' = 2k - y$.

$y' = 2(3) - 1 = 5$.

Answer: $(5, 5)$.

Example 7: Distance preserved, orientation reversed

Explain why a reflection preserves distances between points but reverses orientation.

Solution

Reflecting over the x-axis sends $(x, y) \to (x, -y)$. For two points $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, the horizontal difference $x_2 - x_1$ is unchanged, and the vertical difference becomes $-(y_2 - y_1)$ — same size, opposite sign — so the (squared) distance formula gives an identical result.

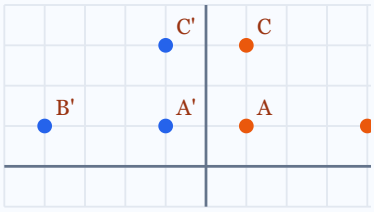
Orientation flips because a mirror image is like viewing a shape through glass: tracing clockwise around the original becomes counterclockwise around the image whenever exactly one coordinate direction is negated.

Answer: Distance survives because squaring erases the sign flip; orientation reverses because exactly one direction is negated.

Example 8: Reflect a whole triangle

$\triangle ABC$ has $A(1, 1)$, $B(4, 1)$, $C(1, 3)$. Reflect over the y-axis. Find all three image vertices.

Solution



Every vertex negates its x-coordinate.

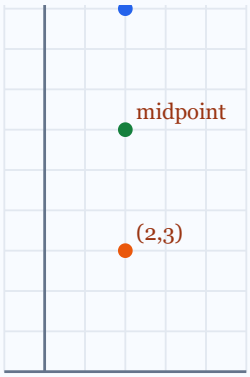
Negate the x-coordinate of each vertex, keeping y the same: $A' = (-1, 1)$; $B' = (-4, 1)$; $C' = (-1, 3)$.

Answer: $A'(-1, 1)$, $B'(-4, 1)$, $C'(-1, 3)$ — the mirror image, same size and shape.

Example 9: Find the line of reflection

A point $(2, 3)$ reflects to $(2, 9)$. Find the equation of the line of reflection.

Solution



The mirror line is the perpendicular bisector of the segment joining a point and its image.

The line of reflection is always the perpendicular bisector of the segment joining a point and its image.

Both points share $x = 2$, so the segment is vertical, making its perpendicular bisector horizontal.

Find the midpoint's y-coordinate: $y = \dfrac{3 + 9}{2} = 6$.

Answer: The line of reflection is $y = 6$.

Example 10: Reflecting twice over the same line

A point is reflected over the x-axis, then that image is reflected over the x-axis again. What is the net result, and why?

Solution

First reflection: $(x, y) \to (x, -y)$. Second reflection over the same line: $(x, -y) \to (x, -(-y)) = (x, y)$.

The two sign flips cancel, returning every point to its original location.

Answer: The net result is the identity transformation — reflecting twice over the *same* line always undoes itself, a preview of composed transformations in Section 3.4.

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: x-axis

Reflect $(4, -2)$ over x-axis.

Answer: **Verification:**

Problem 2: y-axis

Reflect $(-3, 6)$ over y-axis.

Answer: **Verification:**

Problem 3: $y = x$

Reflect $(5, 1)$ over $y = x$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 4: $y = -x$

Reflect $(2, 3)$ over $y = -x$.

Answer: **Verification:**

Problem 5: Vertical

Reflect $(0, 4)$ over $x = 1$.

Answer: **Verification:**

Problem 6: Horizontal

Reflect $(3, 0)$ over $y = 2$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 7: Orientation

Reflection reverses orientation?

Answer: **Verification:**

Problem 8: Congruent?

Figure congruent to its reflection?

Answer: **Verification:**

Problem 9: x-axis

Reflect $(-1, -8)$ over x-axis.

Answer: **Verification:**

Problem 10: y-axis

Reflect $(7, -3)$ over y-axis.

Answer: **Verification:**

Problem 11: Midpoint

Segment joining a point to its image is bisected by what?

Answer: **Verification:**

Problem 12: Over $x = 0$

Reflecting over $x = 0$ is reflecting over which axis?

Answer: **Verification:**

Problem 13: Over $y = 0$

Reflecting over $y = 0$ is reflecting over which axis?

Answer: **Verification:**

Problem 14: Challenge — reflect a triangle

Triangle $P(1, 1)$, $Q(4, 1)$, $R(1, 5)$ is reflected over the y-axis. Find all three image vertices.

Answer: **Verification:**

Problem 15: Challenge — find the mirror line

A point $(5, -2)$ reflects to $(5, 8)$. Find the equation of the line of reflection.

Answer: **Verification:**

✓ Answer Key

1. $(4, 2)$
2. $(3, 6)$
3. $(1, 5)$
4. $(-3, -2)$
5. $(2, 4)$
6. $(3, 4)$
7. Yes
8. Yes
9. $(-1, 8)$
10. $(-7, -3)$
11. The reflection line
12. y-axis
13. x-axis
14. $P'(-1, 1), Q'(-4, 1), R'(-1, 5)$
15. $y = 3$

Great work — keep practising!